MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS)

FOURTH SEMESTER EXAMINATION

(2024 ADMISION ONWARDS)

MG4SECPHY200 Electrical Circuits and Network Skills

Duration: 1.5 hours **Maximum Marks: 50**

Students should attempt at least one question from each course outcome to enhance their overall *outcome attainability.*

**Remember (K), Understand (U), Apply (A), Analyse (An), Evaluate (E), Create (C), Skill (S), Interest (I) and Appreciation (Ap)*

Part A

Multiple Choice Questions

Answer **All** Questions

Each question carries 1 mark

1	The flow of electric charges in a conductor is called: A) Voltage B) Resistance C) Current D) Power	U	1
2	Which device is used to measure electrical power? A) Voltmeter B) Ammeter C) Multimeter D) Wattmeter	An	2
3	In an AC circuit, resistance is replaced by which quantity to calculate current? A) Reactance B) Impedance C) Conductance D) Capacitance	A	3
4	Impedance in an AC circuit depends on A) Resistance only B) Inductance only C) Capacitance only D) Resistance, inductance, and capacitance	U	4
5	Which component allows AC to pass but blocks DC? A) Inductor B) Capacitor C) Resistor D) Battery	A	5
6	What is the primary purpose of a fuse in an electrical circuit? A) To increase the voltage B) To break the circuit during excessive current flow C) To store electrical energy for later use D) To change AC current into DC current	U	6
7	A multimeter can measure: A) Voltage only B) Current only C) Resistance only D) Voltage, current, and resistance	U	1,2
8	If the phase angle between voltage and current is 90°, the power factor of the circuit is A) 1 B) 0.8 C) 0.5 D) 0	An	3

9	An electric motor converts A) electrical energy into heat energy B) mechanical energy into electrical energy C) electrical energy into mechanical energy D) chemical energy into electrical energy	U	4
10	Voltage drop in a cable mainly depends on A) colour of the cable B) length of the cable C) shape of the cable D) insulation colour	A	6

1 * 10= 10

Part B

Short Answer Questions

Answer any **8** Questions

Each question carries 5 marks

11	Distinguish between AC and DC electricity (any two points)	U	1
12	What is a voltmeter? What is it used for? Explain how voltage is measured using a voltmeter with a neat diagram.	A	2
13	Why does reactive power not perform useful work in an AC circuit?	An	3
14	Write about the working of a DC generator	U	4
15	Briefly explain the response of a capacitor to DC and AC sources.	An	5
16	Explain the function of an "Automatic Main Failure" (AMF) switch in a backup power system	U	6
17	State the uses of a multimeter.	U	1,2
18	An AC circuit has an RMS voltage of 200 V and impedance of 20 Ω. Calculate the RMS current. RMS voltage, $V = 200$V Impedance, $Z = 20\Omega$	A	3
19	Differentiate between single-phase, three-phase and DC motors.	An	4
20	Describe different types of cables and their properties	U	6

8 * 5 = 40